MOHAMMED AHSAN

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PROFESSIONAL SUMMARY

DevOps Engineer with **4+ years** of experience designing and automating cloud-native infrastructure on AWS, especially EKS. Skilled in Terraform, GitLab CI, Jenkins, Helm, and GitOps workflows using Argo CD. Focused on building scalable, secure, and automated delivery platforms with strong experience in observability, CI/CD, and infrastructure optimization.

CORE SKILLS

Cloud Platforms & IaC Tools: AWS Cloud, Terraform, Ansible, Linux

Containerization & Orchestration: Docker, Kubernetes, Helm, Amazon EKS

CI/CD & SCM Tools: Jenkins, GitLab, Git, GitHub

Monitoring & Logging: Prometheus, Grafana, ELK, Datadog, CloudWatch

Scripting & Programming: Python, Bash, Go, YAML

AI Tools: GitHub Copilot, Gen AI

PROFESSIONAL EXPERIENCE

Systems Engineer ***Nov' 2024 – Present***

Tata Consultancy Services, Hyderabad

- Designed and deployed AWS infrastructure using Terraform reusable modules (VPC, EKS, RDS, IAM) and implemented a Terratest-based IaC validation framework, reducing provisioning time by 60%, catching ~90% of misconfigurations pre-deployment, and lowering production incidents by 40%.
- Built and managed a production-grade EKS microservices platform using managed node groups, IRSA, HPA/Cluster Autoscaler, and ALB Ingress Controller, supporting 20+ microservices.
- Engineered GitLab CI pipelines with multi-stage build/scan/test workflows, Nexus artifact storage, and environment-specific deployment pipelines with approval gates, ensuring secure production releases.
- Developed and maintained 10+ internal Helm charts with semantic versioning, configurable values, and automated templating workflows, enabling standardized, self-service deployments for multiple app teams.
- Integrated Datadog APM, Logs, and Metrics across EKS and AWS services, establishing full-stack observability with SLO dashboards, proactive alerting and improved MTTR for critical workloads.
- Worked with ArgoCD for GitOps-based deployments and built POC-level implementations to gain hands-on understanding of rolling and blue/green deployment strategies in Kubernetes.
- Worked with application teams to prepare workloads for EKS adoption by building Terraform modules, Helm charts, and CI/CD pipelines required for container-based deployments.

Associate Consultant ***Apr' 2021 – Nov' 2024***

Capgemini Technology Services, Hyderabad

- Engineered Jenkins-based CI/CD pipelines with automated build, testing, quality checks, and controlled deployments, improving reliability and accelerating release cycles.
- Provisioned and automated AWS infrastructure using Terraform, enabling consistent, repeatable, and secure environment deployments across dev/stage/prod.
- Automated configuration management using Ansible, implementing reusable roles and playbooks to streamline application deployments and reduce manual operational effort.
- Implemented centralized monitoring and logging using the ELK Stack, improving system observability and reducing troubleshooting time for application and infrastructure issues.

PROJECTS

Project 1: CI/CD Automation of Java Application on Kubernetes using Jenkins ***GitHub: Link***

Technologies: Jenkins, SonarQube, Nexus, Maven, Kubernetes, Trivy

- Designed and implemented an end-to-end CI/CD pipeline using Jenkins for automated build and deployment of a Java application on Kubernetes.
- Integrated SonarQube for continuous code quality analysis, Nexus for artifact management, and Maven for build automation to ensure high-quality and reliable releases.
- Improved deployment speed and release stability through containerized deployments, gaining hands-on experience with complete DevOps automation workflows.

Project 2: Cloud-Native MERN Stack Deployment on AWS EKS ***GitHub: Link***

Technologies: AWS EKS, ECR, Nginx, Node.js, React, MongoDB, Kubernetes

- Deployed a production-grade MERN stack application on AWS EKS using Docker images stored in Amazon ECR, implementing scalable microservices architecture with Kubernetes.
- Configured StatefulSets, PVCs, Ingress, ConfigMaps, Secrets, health probes, resource limits, and rolling updates, ensuring high availability and reliability of services.
- Troubleshot and resolved real-world issues across networking, containerization, Kubernetes orchestration, and database layers, improving application stability and performance**.**

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate ***Oct' 2024***
- AWS Certified Cloud Practitioner ***Dec' 2023***

EDUCATION

Bachelor of Technology in Computer Science ***Aug' 2016 – Oct' 2020***

Jawaharlal Nehru Technological University, Hyderabad